
Warranty GENE Warehouse Wireframe

A grain-declared structured database over the CDK Drive
and PACCAR warranty extract set — v0.1

PROGRAMME

Hawkins Twin Platform · CDK Twin · Peterbilt Atlantic

DOCUMENT

EgD-HAW-CDK-WH-001 r1

EXTRACT SET

2026-09-17 — 11 source files, 2,991 units, 7,794 repair orders

JOIN SPINE

VIN8 — manufacturer registration to dealer repair order

REPOSITORY

VEglyphDesign/eve-hawkins-cdk-twin · warehouse/

PUBLIC SURFACE

evglyphdesign.github.io/eve-hawkins-cdk-twin/warehouse/

CUSTODY

Schema and rules published · dealer payload never committed

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This directory is the structured database wireframe for the Peterbilt Atlantic warranty lane. It converts a pile of ad-hoc CDK Drive report exports and PACCAR warranty registration reports into one grain-declared star schema with a single join spine, so that every downstream worklist, dashboard and alert reads one shape and no other.

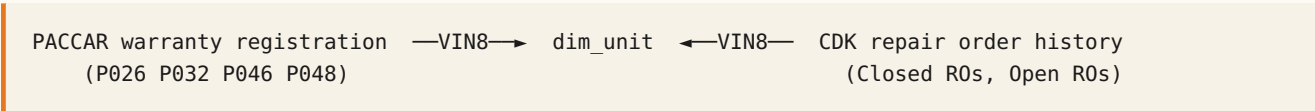
It is a **wireframe**: schema, grain, keys, transforms, marts and data-quality rules.

Custody

No dealer rows are committed to this repository. This repo holds the index, the rules and the schema — never the payload. The loader reads a dealer-owned extract directory and writes the database file outside the repository tree. The only published artefact derived from the data is `profile/2026-09-17-extract-profile.json`, which contains row counts and aggregates and no customer, VIN or contract row.

The join spine — VIN8

The single most important finding in this extract set: **VIN8 joins the manufacturer record to the dealer record.** CDK's `Vehicle ID` on the repair order carries the 8-character VIN suffix, and every PACCAR warranty-registration row is keyed on `VIN8`.



Measured yield on the 2026-09-17 set:

	UNITS
Distinct units in <code>dim_unit</code>	2,991
Warranty-registered (PACCAR side)	1,060
Service-known (CDK RO side)	2,267
Both sides — fully reconciled	336

The secondary key is the customer: CDK's legacy Pick `NAME-FILE` encodes the customer number as `<store_prefix>*<customer_number>`, which resolves 818 of the 821 customers appearing on repair orders back to the customer master. Registered-owner name is *not* a usable join — only 65 of 534 manufacturer-side owner names match a CDK RO customer name exactly, which is exactly why the model joins on `VIN8` and treats names as attributes.

Model

Six dimensions, three facts, two reconciliation aggregates, one quarantine table, six marts.

- Dimensions** — `dim_unit` (VIN8), `dim_customer`, `dim_rooftop`, `dim_warranty_option`, `dim_service_advisor`, `dim_date`.
- Facts** — `fact_warranty_registration` (grain: VIN8 × warranty option, manufacturer truth), `fact_repair_order` (grain: one RO, open and closed unioned under `ro_status`, dealer truth), `fact_service_contract` (grain: one contract, BRP/Torque lane, VIN17-keyed with VIN8 derived).
- Aggregates** — `agg_customer_service_period`, `agg_customer_total_sales`. These are tie-out instruments only. They are never a scoring input; a pre-aggregated total cannot be attributed to a VIN and must not be allowed to look as if it can.

Quarantine — `dq_exception` . Nothing is silently dropped. Every unjoinable vehicle ID, unparseable date and unknown customer lands as a row with a rule code and a severity.

Full ERD in `ERD.md` . DDL in `ddl/wireframe.sql` . Field-level lineage for all 40 mapped columns in `mapping/extract_to_model.csv` .

Marts — the only shapes GENE may read

VIEW	GRAIN	PURPOSE
<code>v_unit_coverage_state</code>	VIN8 × option	coverage window state against a run date, both axes
<code>v_unit_service_profile</code>	VIN8	RO count, last visit, max odometer, lifetime sales
<code>v_rich_target</code>	VIN8	active coverage plus a reachability class
<code>v_whitespace_registered_never_served</code>	VIN8	registered in territory, never in the bays
<code>v_coverage_expiring_180d</code>	VIN8	expiry pressure inside two quarters
<code>v_served_unregistered</code>	VIN8	served here, no manufacturer registration on file

`v_rich_target` classes every unit with live coverage by how the dealership can actually reach it:

REACH CLASS	UNITS	EXPIRING WITHIN 180 DAYS
<code>ON_SITE_NOW</code> — on an open RO today	60	14
<code>KNOWN_CUSTOMER</code> — has RO history here	205	35
<code>REGISTERED_NEVER_SERVICED</code> — registered in territory, never serviced	508	68

What the wireframe already shows

- **773 units carry live manufacturer coverage**, holding 1,802 active coverages — 1,236 engine, 273 aftertreatment, 257 vehicle, plus emissions, electrical, towing and cab-clothing lines.
- **60 units with active coverage are on an open repair order right now**. One is 1 day from time expiry. This is the pre-RO decision window GENE exists to serve.
- **508 registered units have never been serviced at any of the three rooftops**. That is 48% of the manufacturer-registered population, addressable by city because the registration record carries the owner's city.
- **170 of 650 time-active coverages are already past their mileage ceiling**. Coverage is a two-axis test — months *and* miles — and a time-only filter would mis-flag every one of them. `v_unit_coverage_state` therefore exposes `is_time_active` and `miles_remaining_est` as separate columns and refuses to collapse them into one boolean.
- Warranty work is **14.7% of service sales** in the extract window, against \$30.0M customer-pay. The wireframe's job is to move that share deliberately rather than by accident.
- Dealer-side history in scope: 7,534 closed repair orders, \$12.11M total sales, \$5.90M gross profit, 2024-10-01 through 2026-09-16, across three service accounts (`PBNS-S` , `PBDT-S` , `PNBDL-S`) under one accounting account (`PNB-A`).

Boundary

GENE fires **before** the repair order exists. PACCAR's PRWS remains authoritative on claim filing, and nothing in this model submits, adjudicates or authorises a claim. `v_rich_target` produces a prompt for a service writer to decide — a review filter, not a verdict.

Running it

```
python bin/build_wireframe_db.py \  
  --src /dealer-owned/path/to/extracts \  
  --db /dealer-owned/path/out/cdk_wireframe.db \  
  --profile-out profile/$(date +%F)-extract-profile.json
```

The loader is idempotent, rebuilds from scratch each run, and prints an aggregate-only profile. It expects `P026.csv`, `P032.csv`, `P046.csv`, `P048.csv`, `Closed R0s.csv`, `Open R0s.csv`, `Customers.csv`, `service customer sales.csv`, `Customer total sales.csv`, `Advantage plus customers.csv` and `Canada General contracts.csv`. Missing files are recorded as `SOURCE_MISSING` warnings rather than failing the run, because the extract set arrives in pieces.

SQLite is the wireframe target because it is inspectable with no server. The DDL is deliberately portable to the Azure PostgreSQL landing zone that the CDK Twin ingestion pipeline targets; see [DATA-QUALITY.md](#) for the promotion gates that must pass before a Postgres load is trustworthy.

Data Quality — 2026-09-17 extract set

Nothing is silently dropped. Every rejection lands in `dq_exception` with a rule code and a severity, and the counts below are reproducible by rerunning the loader.

RULE	SEVERITY	COUNT	MEANING
VIN8_LENGTH	WARN	1,223	RO rows whose <code>Vehicle ID</code> is not 8 or 17 characters, so they cannot reach the warranty registration
DATE_PARSE	WARN	213	PACCAR registration rows with a missing or unparseable in-service date — coverage end cannot be computed
RO_DUPLICATE	WARN	12	an RO number present in both the open and closed extract; the closed row wins
CUST_UNKNOWN	INFO	3	a customer on an RO absent from the customer master; a stub row is created

The five findings that constrain the model

- 1. Vehicle ID is a VIN8, not a VIN.** 2,141 of 2,436 distinct CDK vehicle IDs are 8 characters, 245 are 6, and 4 are full 17-character VINs. The 6-character values cannot be lengthened by inference and must not be fuzzy-matched to a VIN8 — a wrong VIN match would attach one fleet's coverage to another fleet's truck. They are quarantined, counted, and excluded from `v_rich_target`, which is why the mart population is smaller than the RO population.
- 2. The four PACCAR reports are disjoint partitions, not overlapping views.** P026 (587 VINs), P032 (273), P046 (196) and P048 (4) share zero VIN8 between them, and their city profiles differ — Fredericton and Dieppe in P026, Summerside and Cap Pelé in P032, Kentville and Truro in P046, Quebec towns in P048. They union to exactly 1,060 units, which matches the count of `BASE/VEH` coverage rows exactly: one base vehicle warranty per unit. Loading them as one union is therefore correct, and any attempt to reconcile them against each other is wasted work.
- 3. Coverage expiry has two axes and the mileage axis is already binding.** Of 650 time-active coverages where an odometer reading exists, 170 are already past their mileage ceiling and 26 are within 25,000 of it. A time-only filter would present 170 dead coverages as live. `v_unit_coverage_state` exposes `is_time_active` and `miles_remaining_est` separately and never collapses them.
- 4. Owner name is not a join key.** Only 65 of 534 manufacturer-side registered-owner names match a CDK RO customer name exactly. Registration owner is stored as an attribute of the unit and is used for outreach copy, never for identity resolution.
- 5. The aggregate extracts are not a population.** `Customer total sales.csv` is a top-100 ranked report and `service customer sales.csv` covers 2,693 customers against 16,022 in the master. They are loaded into `agg_*` tables for tie-out and are barred from scoring, because a customer-period total cannot be attributed to a VIN.

Promotion gates — before any Azure PostgreSQL load is trusted

- `dq_exception` carries zero `BLOCK` rows.
- `SUM(total_sales) over ro_status='CLOSED'` ties to the closed-RO extract total (\$12,105,138.15 on this set, 7,534 ROs).
- `COUNT(*) in fact_warranty_registration` equals the summed source report rows less `VIN8_LENGTH/BLOCK` rejections (3,258 on this set).
- Every `fact_repair_order.vin8` that is non-null is 8 characters.
- `v_rich_target` contains no row whose nearest expiry is negative.
- `dim_unit` row count equals the distinct union of registered and serviced VIN8 (2,991 on this set).

Known gaps in the extract set

- No RO **line detail** — the extracts are RO headers only. Labour operation codes, SRT hours, part numbers and the 3-C complaint text are absent, so margin estimate and 3-C compliance cannot yet be scored per job. This is the single highest-value addition to the next extract request.
 - No claim-side data on the Peterbilt lane. PRWS claim history is not in this set, so booked-versus-eligible cannot be closed on the truck side yet.
 - No SmartLINQ / PACCAR Solutions fault-code feed, so the event-driven rung is absent.
 - Odometer arrives only via ROs, so units never serviced here have no mileage reading and their mileage axis is unknown — reported as `NULL` , never as zero.
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Pour le bien-être du peuple.